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Fast inference for quantile regression with tens of millions of observations☆

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ABSTRACT

Big data analytics has opened new avenues in economic research, but the challenge of analyzing datasets with tens of millions of observations is substantial. Conventional econometric methods based on extreme estimators require large amounts of computing resources and memory, which are often not readily available. In this paper, we focus on linear quantile regression applied to "ultra-large" datasets, such as U.S. decennial censuses. A fast inference framework is presented, utilizing stochastic subgradient descent (S-subGD) updates. The inference procedure handles cross-sectional data sequentially: (i) updating the parameter estimate with each incoming "new observation", (ii) aggregating it as a *Polyak–Ruppert* average, and (iii) computing a pivotal statistic for inference using only a solution path. The methodology draws from time-series regression to create an asymptotically pivotal statistic through random scaling. Our proposed test statistic is calculated in a fully online fashion and critical values are calculated without resampling. We conduct extensive numerical studies to showcase the computational merits of our proposed inference. For inference problems as large as $(n, d) \sim (10^7, 10^3)$, where n is the sample size and d is the number of regressors, our method generates new insights, surpassing current inference methods in computation. Our method specifically reveals trends in the gender gap in the U.S. college wage premium using millions of observations, while controlling over 10^3 covariates to mitigate confounding effects.

1. Introduction

Quantile regression (QR) has been increasingly popular in economics since the groundbreaking work of Koenker and Bassett (1978). Many significant developments have been made to this important statistical methodology. Meanwhile, economists are now able to analyze datasets that contain tens of millions of observations. For instance, one of the important economic applications of quantile regression is to study the wage structure. The 5% sample in the 2000 U.S. Census contains more than 14 million observations

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and more than 4 million even after restricting the sample to a subpopulation of working White adults. One motivation of using datasets with such "ultra-large" sample sizes is to deal with econometric models with a large number of parameters within the framework of simple parametric models. Suppose that there are d parameters to estimate with a sample of size n. A parametric regression model can be as large as $d \sim 1,000$ because a large number of controls might be necessary. In the setting of "ultra-large" sample size with $n \sim 10^7$, straightforward parametric estimators can be used with $d \sim 1,000$ under standard textbook assumptions without demanding conditions such as sparsity assumptions.[1]

With datasets containing tens of millions of observations, however, one major challenge of QR is that it would require huge computing powers and memory that are often not accessible using ordinary personal computers. While inference is desirable in empirical studies to quantify statistical uncertainty, it often demands more computational resources than point estimation. Typically, asymptotic normal inference with QR estimators involves solving optimization problems and computing asymptotic covariance matrices. Computationally efficient implementation of both tasks is crucial with ultra-large datasets.

In this paper, we focus on this challenge, by studying the QR inference problem in the context of $n \sim 10^7$. We propose a fast statistical inference framework to analyze cross-sectional data with millions of observations, a typical sample size for the census data, embracing stochastic gradient descent techniques, one of the most active research areas in machine learning. This literature dates back to Robbins and Monro (1951). While we focus on the QR framework, whose inference has been well known to be a hard problem, our proposed method can be naturally generalized to other econometric frameworks.

Empirically, we use the data from IPUMS USA (Ruggles et al., 2022), a common data source for the U.S. wage structure, to study the trends in the gender gap in terms of the college wage premium. In our empirical application, we aim to control for work experience by flexibly interacting workers' age with gender and state fixed effects, which creates over one thousand regressors, thereby motivating the need of ultra-large datasets. We find that existing inference methods are not applicable due to time or memory constraints, while our inference method provides new insights into trends in the gender gap. There has been a large literature on college wage premium and wage structure. See, for instance, Chamberlain (1994), Buchinsky (1994, 1998), Gosling et al. (2000), Angrist et al. (2006), Chernozhukov et al. (2013) among others. To the best of our knowledge, this empirical illustration is the first to obtain confidence intervals based on quantile regression with more than 4 million observations and more than 1000 regressors.

1.1. Standard inference for quantile regression

Consider the setting of a linear quantile regression model for which data $\{Y_i \equiv (y_i, x_i) \in \mathbb{R}^{1+d} : i = 1, \dots, n\}$ are generated from

$$y_i = x_i'\beta^* + \varepsilon_i, \tag{1}$$

where β^* is a vector of unknown parameters and the unobserved error ε_i satisfies $P(\varepsilon_i \le 0|x_i) = \tau$ for a fixed quantile $\tau \in (0,1)$. Note that β^* is characterized by

$$\beta^* := \arg\min_{\beta\in\mathbb{R}^d} Q(\beta),$$

where $Q(\beta) := \mathbb{E}[q(\beta, Y_i)]$ with the check function $q(\beta, Y_i) := (y_i - x_i'\beta)(\tau - I\{y_i - x_i'\beta \le 0\})$. Here, $I(\cdot)$ is the usual indicator function. The standard approach for estimating β^* is to use the following M-estimator originally proposed by Koenker and Bassett (1978):

$$\hat{\beta}_n := \arg\min_{\beta\in\mathbb{R}^d} \frac{1}{n}\sum_{i=1}^n q(\beta, Y_i). \tag{2}$$

See Koenker (2005) and Koenker (2017) for a monograph and a review of recent developments, respectively.

There have been two potential challenges for the inference of standard quantile regression. The first is the computational problem. The optimization problem is typically reformulated to a linear programming problem, and solved using interior-point algorithms (e.g., Portnoy and Koenker, 1997). The second is associated with statistical inference. As was shown by Koenker and Bassett (1978), the asymptotic distribution of the M estimator is as follows:

$$\sqrt{n}(\hat{\beta}_n - \beta^*) \xrightarrow{d} N(0, \tau(1-\tau)H^{-1}\mathbb{E}[x_i x_i']H^{-1}), \quad H = \mathbb{E}[f_\varepsilon(0|x_i)x_i x_i'], \tag{3}$$

where $f_\varepsilon(\cdot|x_i)$ is the conditional distribution of ε_i given x_i, assuming data are independent and identically distributed (i.i.d.). Hence, in the heteroskedastic setting, standard inference based on the "plug-in" method would require estimating matrix H, which involves the conditional density function $f_\varepsilon(\cdot|x_i)$. An alternative inference would be based on bootstrap.

One of the attempts to solving the computational/inference difficulties is to rely on the *smoothing* idea, via either smoothing the estimating equation (Horowitz, 1998) or the convolution-type smoothing (Fernandes et al., 2021; Tan et al., 2022; He et al., 2023). Both require a choice of the smoothing bandwidth. In particular, the convolution-type smoothing (`conquer` as coined by He et al. (2023)) has received recent attention in the literature because the optimization problem is convex, so it is more scalable. Figure 1 of He et al. (2023) shows that `conquer` works well for point estimations when the sample size ranges from $10^3 \sim 10^6$ with the number of regressors $d \approx n^{1/2}$.

Meanwhile, in terms of inference, both types of smoothed estimators are first-order asymptotically equivalent to the unsmoothed estimator $\hat{\beta}_n$, and the scale of the numerical studies of these solutions is less ambitious. For instance, the largest model considered for inference in He et al. (2023) is merely $(n, d) = (4000, 100)$ (see Figure 7 in their paper). Therefore, there is a scalability gap between point estimation and inference in the literature. In this paper, we aim to bridge this gap by proposing a method for fast inference.

[1] A classical condition for a uniform normal approximation of an M-estimator is $(d \log n)^{3/2}/n \to 0$ (Portnoy, 1985).

1.2. The proposed fast S-subGD framework

We focus on the inference problem at the scale up to $(n, d) \sim (10^7, 10^3)$, which we refer to as an "ultra-large" quantile regression problem. This is a possible scale with IPUMS USA, but very difficult to deal with using benchmark inference procedures for quantile regression. To tackle this large-scale problem, we estimate β^* via stochastic (sub)gradient descent (S-subGD). Our assumption on the asymptotic regime is that d is fixed but n grows to infinity.

Suppose that we have i.i.d. data from a large cross-sectional survey:

$$Y_1, \ldots, Y_n, \quad Y_i = (x_i, y_i),$$

where the ordering of these observations is randomized. Our method produces a sequence of estimators, denoted by β_i, which is a solution path being updated as

$$\beta_i = \beta_{i-1} - \gamma_i \nabla q\left(\beta_{i-1}, Y_i\right).$$

Here Y_i is a new observation in the randomized data sequence. Also, $\nabla q\left(\beta_{i-1}, Y_i\right)$ is a subgradient of the check function with respect to the current update, and γ_i is a predetermined learning rate. Then, we take the average of the sequence

$$\bar{\beta}_n := \frac{1}{n}\sum_{i=1}^{n} \beta_i,$$

which is called as the Polyak (1990)-Ruppert (1988) average in the machine learning literature. It will be shown in this paper that $\sqrt{n}(\bar{\beta}_n - \beta^*)$ is asymptotically normal with the asymptotic variance the same as that of the standard Koenker and Bassett (1978) estimator $\widehat{\beta}_n$. Therefore, it achieves the same first-order asymptotic efficiency.

The main novelty of this paper comes from how to conduct inference based on $\bar{\beta}_n$. We use a sequential transformation of β_i's in a suitable way for on-line updating, and construct asymptotically pivotal statistics. That is, we studentize $\sqrt{n}\left(\bar{\beta}_n - \beta^*\right)$ via a random scaling matrix $\widehat{V}_n$ whose exact form will be given later. The resulting statistic is not asymptotically normal but *asymptotically pivotal* in the sense that its asymptotic distribution is free of any unknown nuisance parameters; thus, its critical values are easily available. Furthermore, the random scaling quantity $\widehat{V}_n$ does not require any additional input other than stochastic subgradient path β_i, and can be computed fast.

The main contribution of this paper is computational. We combine the idea of stochastic subgradient descent with random scaling and make large-scale inference for quantile regression much more practical. We provide a discussion of computational complexity and show that our algorithm is comparable to the existing ones in terms of point estimation and is superior to those in terms of inference (especially, subvector inference). See Section 2.6 for details. In addition, we demonstrate the usefulness of our method via Monte Carlo experiments. Empirically, we apply it to studying the gender gap in college wage premiums using the IPUMS dataset. The proposed inference method of quantile regression to the ultra-big dataset reveals some interesting new features. First, it shows heterogeneous effects over different quantile levels. Second, we find that the female college wage premium is significantly higher than that of male workers at the median, which is different from what have been concluded in the literature. In fact, although the S-subGD inference tends to be conservative than the standard normal approximation, it reveals statistically significant results while controlling over 10^3 covariates to mitigate confounding effects. With more availability of such a large dataset, the S-subGD method will make it possible to obtain convincing empirical evidence for other analyses. Another topic is to examine the relationship between seasonality and birth weight (e.g., Buckles and Hungerman, 2013; Currie and Schwandt, 2013; Chatterji et al., 2022) because the quantiles of birth weight are of interest and birth data from vital statistics contain tens of millions of observations.[2]

1.3. The related literature

Our estimator is motivated from the literature on *online learning*, where data were collected in a streaming fashion, and as a new observation Y_i arrives, the estimator is updated to β_i. Under the M-estimation framework with strongly convex, differentiable loss functions, Polyak and Juditsky (1992) showed that the Polyak (1990)-Ruppert (1988) average is asymptotically normal. More recently, Fang et al. (2018), Chen et al. (2020), and Zhu et al. (2023) studied the statistical inference problem and proposed methods that require the implementation of bootstrap, consistent estimation of the asymptotic variance matrix, or the use of "batch-means" approach, respectively. Besides the smoothness assumption that excludes quantile regression, neither consistent estimation of the asymptotic variance matrix nor implementation of bootstrap is computationally attractive for datasets as large as those with which this paper is concerned. The batch-means approach does not seem to provide adequate finite sample coverage even if n is sufficiently large (see, e.g., Lee et al., 2022, in the context of linear mean regression and logistic regression models).

The idea of random scaling is borrowed from the time-series literature on fixed bandwidth heteroskedasticity and autocorrelation robust (HAR) inference (e.g., Kiefer et al., 2000; Sun et al., 2008; Sun, 2014; Lazarus et al., 2018), and it has been recently adopted in the context of stochastic gradient descent (SGD) in machine-learning problems: online inference for linear mean regression (Lee et al., 2022), federated learning (Li et al., 2022), and Kiefer–Wolfowitz methods (Chen et al., 2021) among others.[3] The current paper shares the basic inference idea with our previous work (Lee et al., 2022); however, quantile regression is sufficiently different

[2] For example, the National Bureau of Economic Research has public use data at https://www.nber.org/research/data/vital-statistics-natality-birth-data.

[3] Li et al. (2022) and Chen et al. (2021) applied the idea of random scaling after our previous work (Lee et al., 2022) appeared on arXiv.

from mean regression and it requires further theoretical development that is not covered in the existing work. In particular, our previous work is based on Polyak and Juditsky (1992), which limits its analysis to differentiable and strongly convex objective functions; however, the check function $q(\beta, Y_i)$ for quantile regression is non-differentiable and is not strongly convex. We overcome these difficulties by using the results given in Gadat and Panloup (2023). One could have adopted convolution-type smoothing for S-subGD as the corresponding objective function is differentiable and locally strongly convex. We do not go down this alternative route in this paper as the existence of a sequence of smoothing bandwidths converging to zero complicates theoretical analysis and it would be difficult to optimally choose the sequence of smoothing bandwidths along with the sequence of learning rates. Previously, Pesme and Flammarion (2020) studied online robust regression in the context of a Gaussian linear model with an adversarial noise; in fact, their algorithm corresponds to our median regression case (see Eq. (2) in their paper) but they did not study quantile regression models. Gadat and Panloup (2023) considered, as their example, the recursive quantile estimator, which corresponds to the quantile regression estimator with an intercept being only the regressor. Neither of these two papers delved into the issue of inference.

In the econometric literature, Forneron and Ng (2021) and Forneron (2022) developed SGD-based resampling schemes that deliver both point estimates and standard errors within the same optimization framework. The major difference is that their stochastic path is a Newton–Raphson type which requires computing the Hessian matrix or its approximation. Hence, their framework is not applicable to quantile regression because the check function is not twice differentiable. In other words, for quantile regression inferences with over millions of observations, estimating the inverse Hessian matrix H^{-1} in (3) is a task that we particularly would like to avoid.

As an alternative to full sample estimation, one may consider sketching (e.g., see Lee and Ng, 2020, for a review from an econometric perspective). For example, Portnoy and Koenker (1997) developed a preprocessing algorithm that can be viewed as a sketching method and Yang et al. (2013) proposed a fast randomized algorithm for large-scale quantile regression and solved the problem of size $n \sim 10^{10}$ and $d = 12$ by randomly creating a subsample of about $n = 10^5$. However, the theoretical analysis carried out in Yang et al. (2013) is limited to approximations of the optimal value of the check function and does not cover the issue of inference. As an alternative sketching method, one may employ Wang and Ma (2021) and construct a random subsample using data-dependent, nonuniform weights to improve the efficiency of the estimator. In mean regression models, the precision of a sketched estimator depends on the subsample size and is typically worse than that of the full sample estimator (e.g., Lee and Ng, 2020, 2022). We expect a similar phenomenon for quantile regression models.

1.4. Notation

Let a' and A', respectively, denote the transpose of vector a and matrix A. Let $\|a\|$ denote the Euclidean norm of vector a and $\|A\|$ the Frobenius norm of matrix A. Also, let $\ell^\infty[0,1]$ denote the set of bounded continuous functions on $[0,1]$. Let $I(A)$ be the usual indicator function, that is $I(A) = 1$ if A is true and 0 otherwise. For a symmetric, positive definite matrix S, let $\lambda_{\min}(S)$ denote its smallest eigenvalue.

2. A fast algorithm for quantile inference

2.1. Stochastic subgradient descent (S-subGD)

In this section, we propose our inference algorithm. Suppose that we have cross-sectional data $\{Y_1, \dots, Y_n\}$, where $Y_i = (x_i, y_i)$. First, we randomize the ordering of the observed data:

$$Y_1 = (x_1, y_1), \dots, Y_n = (x_n, y_n).$$

We start with an initial estimator, denoted by β_0, using a method which we shall discuss later. Then produce an S-subGD solution path which updates according to the rule:

$$\beta_i = \beta_{i-1} - \gamma_i \nabla q\left(\beta_{i-1}, Y_i\right), \tag{4}$$

where the updating subgradient depends on a *single "next" observation* Y_i:

$$\nabla q(\beta, Y_i) := x_i[I\{y_i \le x_i'\beta\} - \tau]. \tag{5}$$

This is the subgradient of $q\left(\beta, Y_i\right)$ with respect to β. Here β_{i-1} is the "current" estimate, which is updated to β_i when the next observation Y_i comes into play. Also γ_i is a pre-determined step size, which is also called a learning rate and is assumed to have the form $\gamma_i := \gamma_0 i^{-a}$ for some constants $\gamma_0 > 0$ and $a \in (1/2, 1)$. Then, we take the average of the sequence $\bar{\beta}_n := \frac{1}{n}\sum_{i=1}^n \beta_i$ as the final estimator.

Computing this estimator does not require storing the historical data; thus, it is very fast and memory-efficient even when $n \sim 10^7$. In fact, as we shall see below, the computational gain is much more substantial for inference.

2.2. The pivotal statistic

It is most common to make inference based on a consistent estimator of the asymptotic variance for $\sqrt{n}(\bar{\beta}_n - \beta^*)$, which is computationally demanding. For instance, one needs to estimate the inverse Hessian matrix H^{-1} in (3). Instead of pursuing

Table 1
Asymptotic critical values of the t-statistic.
Source: Abadir and Paruolo (1997, Table I).

Probability	90%	95%	97.5%	99%
Critical Value	3.875	5.323	6.747	8.613

Note. The table gives one-sided asymptotic critical values that satisfy $\Pr(\hat{t} \le c) = p$ asymptotically, where $p \in \{0.9, 0.95, 0.975, 0.99\}$.

asymptotic normal inference using a consistent estimator of the asymptotic variance, we employ asymptotic mixed-normal inference based on a suitable random scaling method through the partial sum process of the solution path as done in the *fixed-b* standardization. Motivated by the fact that the solution path β_i is sequentially updated, we apply the random scaling approach by defining:

$$\widehat{V}_n := \frac{1}{n}\sum_{s=1}^{n}\left\{\frac{1}{\sqrt{n}}\sum_{i=1}^{s}\left(\beta_i - \bar{\beta}_n\right)\right\}\left\{\frac{1}{\sqrt{n}}\sum_{i=1}^{s}\left(\beta_i - \bar{\beta}_n\right)\right\}'. \tag{6}$$

Inference will be conducted based on the standardization using $\widehat{V}_n$. Computing $\widehat{V}_n$ can be efficient even if $n \sim 10^7$ or larger, since it can be sequentially computed as detailed later. Furthermore, its subvector inference involves only the associated elements, substantially reducing the computational burden.

Once $\bar{\beta}_n$ and $\widehat{V}_n$ are obtained, the inference is straightforward. For example, for the j th component of $\bar{\beta}_n$, let $\widehat{V}_{n,jj}$ denote the (j,j) th diagonal entry of $\widehat{V}_n$. The t-statistic is then

$$\frac{\sqrt{n}\left(\bar{\beta}_{n,j} - \beta^*_j\right)}{\sqrt{\widehat{V}_{n,jj}}}, \tag{7}$$

whose asymptotic distribution is mixed normal and symmetric around zero. We will formally derive it in the next section. The mixed normal asymptotic distribution for the t-statistic in (7) is the same as the distribution of the statistics observed in the estimation of the cointegration vector by Johansen (1991) and Abadir and Paruolo (1997) and in the heteroskedasticity and autocorrelation robust inference in Kiefer et al. (2000). They are different statistics but converge to the identical distribution, a functional of the standard Wiener process as shown by Abadir and Paruolo (2002). We can use the t-statistic to construct the $(1-\alpha)$ asymptotic confidence interval for the jth element β^*_j of β^* by

$$\left[\bar{\beta}_{n,j} - \mathrm{cv}(1-\alpha/2)\sqrt{\frac{\widehat{V}_{n,jj}}{n}},\ \bar{\beta}_{n,j} + \mathrm{cv}(1-\alpha/2)\sqrt{\frac{\widehat{V}_{n,jj}}{n}}\right],$$

where the critical value $\mathrm{cv}(1-\alpha/2)$ is tabulated in Abadir and Paruolo (1997, Table I). For easy reference, the critical values are reproduced in Table 1. For instance, the critical value for $\alpha = 0.05$ is 6.747. Critical values for testing linear restrictions $H_0 : R\beta^* = c$ can be found in Kiefer et al. (2000, Table II).

2.3. Practical details for memory efficiency

While computing $(\widehat{V}_n, \bar{\beta}_n)$ is straightforward for datasets of medium sample sizes, it is not practical when $n \sim 10^7$ or even larger size, as storing the entire solution path β_i can be infeasible for memory of usual personal computing devices.

We can construct a solution path for $(\widehat{V}_n, \bar{\beta}_n)$ along the way of updating β_i, so all relevant quantities of the proposed inference can be obtained sequentially. Specifically, let $(\bar{\beta}_{i-1}, \beta_{i-1})$ be the current update when we use up to $i-1$ observations. We then update using the new observation Y_i by:

$$\beta_i = \beta_{i-1} - \gamma_i \nabla q(\beta_{i-1}, Y_i), \tag{8}$$

$$\bar{\beta}_i = \bar{\beta}_{i-1}\frac{i-1}{i} + \beta_i \frac{1}{i}, \tag{9}$$

$$\widehat{V}_i = i^{-2}\left(A_i - \bar{\beta}_i b_i' - b_i \bar{\beta}_i' + \bar{\beta}_i \bar{\beta}_i' \sum_{s=1}^{i} s^2\right), \tag{10}$$

where the intermediate quantities A_i and b_i are also updated sequentially:

$$A_i = A_{i-1} + i^2 \bar{\beta}_i \bar{\beta}_i', \quad b_i = b_{i-1} + i^2 \bar{\beta}_i.$$

Therefore computing $(\bar{\beta}_n, \widehat{V}_n)$ can be cast sequentially until $i = n$, and does not require saving the entire solution path of β_i.

A good initial value could help achieve the computational stability in practice, though it does not matter in the theoretical results. We recommend applying `conquer` in He et al. (2023) to the subsample (e.g. 5% or 10%) to estimate the initial value, which we adopted in our numerical experiments. Specifically, we start with a smooth quantile regression proposed in Fernandes et al. (2021) and He et al. (2023):

$$\beta_0 = \arg\min \sum_{i \in S} q_n(\beta, Y_i),$$

where q_n is a convolution-type-smoothed checked function, and $\mathcal{S}$ is a randomized subsample. The conquer algorithm is very fast to compute a point estimate of β_0 but it is much less scalable with respect to inference (see the Monte Carlo results later in the paper). For the intermediate quantities A_i and b_i, we set $A_0 = 0$ and $b_0 = 0$.

Another approach is the burn-in method that applies the S-subGD to some initial observations, and throw them away for the main inference procedure. In our experiments, initial values estimated by conquer outperform those by the burn-in method because conquer tends to produce high-quality initial solutions.

2.4. Choice of the learning rate

We introduce a practical guideline for determining the value of $\gamma_i = \gamma_0 i^{-a}$. Let $\hat{\sigma}$ denote the sample standard deviation of y_i using subsample $\mathcal{S}$. Then, we suggest to fix $a = 0.501$ and select the initial learning rate γ_0 via

$$\gamma_0 = \frac{1}{\hat{\sigma}} \frac{\phi(\Phi^{-1}(\tau))}{\sqrt{\tau(1-\tau)}}, \tag{11}$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are standard normal density and distribution functions, respectively. For example, if $\tau = 0.5$, then $\gamma_0 \approx 0.798/\hat{\sigma}$; if $\tau = 0.1$ or 0.9, then $\gamma_0 \approx 0.585/\hat{\sigma}$. The rule-of-thumb selection of γ_0 is heuristically motivated from the asymptotic distribution of the sample quantile when the regression errors are normally distributed.

2.5. Subvector inference

In most empirical studies, while many covariates are being controlled in the model, of interest are often just one or two key independent variables that are policy-related. In such cases, we are particularly interested in making inference for subvectors, say β_1^*, of the original vector β^*. Subvector inference is not often direct target of interest in the quantile regression literature, partially because inference regarding the full vector β^* and then converting to the subvector is often scalable up to the medium sample size. But this is no longer the case for the ultra-large sample size that is being considered here.

It is straightforward to tailor our updating algorithm to focusing on subvectors. Continue denoting β_i as the ith update of the full vector. While the full vector β_i still needs to be computed in the S-subGD, both the Polyak–Ruppert average and the random scale can be updated only up to the scale of the subvector. Specifically, let $\bar{\beta}_{i,sub}$ denote the subvector of $\bar{\beta}_i$, corresponding to the subvector of interest. Also, let $\widehat{V}_{i,sub}$ denote the sub-matrix of $\widehat{V}_i$; both are updated according to the following rule:

$$\beta_i = \text{updated the same as (8)}, \tag{12}$$

$$\bar{\beta}_{i,sub} = \bar{\beta}_{i-1,sub}\frac{i-1}{i} + \beta_{i,sub}\frac{1}{i}, \tag{13}$$

$$\widehat{V}_{i,sub} = i^{-2}\left(A_{i,sub} - \bar{\beta}_{i,sub}b'_{i,sub} - b_{i,sub}\bar{\beta}'_{i,sub} + \bar{\beta}_{i,sub}\bar{\beta}'_{i,sub}\sum_{s=1}^{i} s^2\right), \tag{14}$$

$$A_{i,sub} = A_{i-1,sub} + i^2\bar{\beta}_{i,sub}\bar{\beta}'_{i,sub}, \quad b_{i,sub} = b_{i-1,sub} + i^2\bar{\beta}_{i,sub}. \tag{15}$$

In most cases, steps (13)–(15) only involve small dimensional objects.

2.6. Computational complexity

Portnoy and Koenker (1997) demonstrated that the computational complexity of the interior point algorithm for quantile regression can be on the order of $\mathcal{O}(nd^3\log^2 n)$ (see discussions after Theorem 5.1 of Portnoy and Koenker (1997)). They further proposed an initial phase of preprocessing that effectively reduces computational complexity to $\mathcal{O}(n^{2/3}d^3\log^2 n) + \mathcal{O}(nd)$ (see Portnoy and Koenker, 1997, p. 290). The basic idea behind preprocessing is that using a preliminary estimate of β^*, one can remove observations that are above and below the quantile regression line with high probability and add two pseudo-observations called "globs", resulting in the reduction of the sample size from n to $\mathcal{O}(n^{2/3})$.[4] These results indicate that the interior point algorithm works well with a large n but not with a large d. In addition, Chen (2007) developed a smoothing algorithm that has the computational complexity of $\mathcal{O}(nd^{2.5})$, which reduces the dependence on d from d^3 to $d^{2.5}$. In He et al. (2023), conquer is computed using gradient descent (GD), which has the computational complexity of $\mathcal{O}(ndk)$, where k is the number of iterations in GD.

Table 2 summarizes the computational cost of estimation and inference. Importantly, the computational gain of our stochastic subgradient method is *not* in terms of point estimation, but of inference. Specifically, our S-subGD method has the computational complexity of $\mathcal{O}(nd)$. If the GD algorithm of conquer converges fast in a finite number of iterations, which seems to be the case empirically, we expect that our algorithm will be comparable—will not be superior—to conquer in terms of point estimation. However, note that GD method can suffer from memory constraints when nd is very large.

However, the comparative advantage is more pronounced in statistical inference, which incurs additional computational costs. Our inference, based on random scaling, entails an additional cost of order $\mathcal{O}(nd^2)$. Nevertheless, it reduces to $\mathcal{O}(nd)$ when the object of inference is a subvector of β^*, with a dimension of $s = \mathcal{O}(d^{1/2})$. This reduction occurs because we only need to update a submatrix

[4] It is an interesting future research topic to examine whether and how preprocessing can be used in our inference procedure.

Table 2
Computational complexity for different methods.

Computational Cost for Estimation	
QR (interior point)	$\widetilde{\mathcal{O}}(nd^3)$
QR (interior point with preprocessing)	$\widetilde{\mathcal{O}}(n^{2/3}d^3) + \mathcal{O}(nd)$
CONQUER (GD)	$\mathcal{O}(ndk)$
S-subGD	$\mathcal{O}(nd)$
Additional Computational Cost for Inference	
QR (plugin)	$\mathcal{O}(nd^2)$
QR (bootstrap with preprocessing)	$\widetilde{\mathcal{O}}(bn^{2/3}d^3) + \mathcal{O}(bnd)$
CONQUER (plugin)	$\mathcal{O}(nd^2)$
CONQUER (bootstrap)	$\mathcal{O}(bndk)$
S-subGD (online bootstrap)	$\mathcal{O}(bnd)$
S-subGD (random scaling)	$\mathcal{O}(nd^2)$
S-subGD (subvector random scaling)	$\mathcal{O}(ns^2)$
S-subGD (diagonal random scaling)	$\mathcal{O}(nd)$

Notes. The symbol $\widetilde{\mathcal{O}}(\cdot)$ does not explicitly show logarithmic factors. Here, n is the sample size; d is the dimension of regressors; k is the number of iterations in CONQUER; b is the number of bootstrap draws; s is the dimension of the subvector of interest.

of the random scaling matrix, as discussed in the previous section. Similar gains are achieved when focusing on the inference of each element separately, as is common in applied practice. This is again because we only need to update the diagonal elements, referred to as diagonal random scaling in Table 2. We emphasize that achieving the same order of computational cost for subvector inference is generally not possible with plug-in inference. This is due to the fact that the plug-in estimator of the sandwich formula ($H^{-1}\Sigma H^{-1}$) of asymptotic variance in (3) cannot be computed using only diagonal elements or a submatrix.

As for the computational cost of competing methods for QR inference, first, the bootstrap inference will be time consuming as the computational cost will be a multiple of bootstrap draws. For instance, bootstrap inference with `conquer` has the computational complexity of $\mathcal{O}(bndk)$, where b is the number of bootstrap draws. Second, it requires access to the entire sample to conduct standard plug-in inference based on asymptotic normality. Thus, the plug-in inference has the additional computational complexity of $\mathcal{O}(nd^2)$ even with the `conquer` algorithm. This will be faster than bootstrap inference if $k > d$, but cannot be reduced to $\mathcal{O}(ns^2)$ or $\mathcal{O}(nd)$ even for subvector inference.

In short, our algorithm can be as competitive as `conquer` in terms of computational cost, while offering the advantages of lower memory requirements and faster computations for subvector inference.

3. Asymptotic theory

In this section, we present asymptotic theory that underpins our inference method. We consider the following QR model:

$$y_i = x_i'\beta_\tau^* + \varepsilon_i,$$

where $P(\varepsilon_i \le 0|x_i) = \tau$ for a fixed quantile $\tau \in (0,1)$. Denote by $\beta^* = \beta_\tau^*$ for simplicity.

Assumption 3.1 (*Conditions for Quantile Regression*)**.** Suppose:

(i) The data $\{Y_i \equiv (y_i, x_i)\}_{i=1}^n$ are independent and identically distributed.
(ii) The conditional density $f_\varepsilon(\cdot|x_i)$ of ε_i given x_i and its partial derivative $\frac{d}{d\varepsilon}f_\varepsilon(\cdot|x_i)$ exist, and furthermore, $\sup_b \mathbb{E}[\|x_i\|^3 A(b,x_i)] < C$ for some constant $C < \infty$, where

$$A(b,x_i) := \left|\frac{d}{d\varepsilon}f_\varepsilon(x_i'b|x_i)\right| + f_\varepsilon(x_i'b|x_i).$$

(iii) There exist positive constants ϵ and c_0 such that

$$\inf_{|\beta-\beta^*|<\epsilon} \lambda_{\min}\left(\mathbb{E}[x_i x_i' f_\varepsilon(x_i'(\beta-\beta^*)|x_i)]\right) > c_0.$$

(iv) $\mathbb{E}[(\|x_i\|^6 + 1)\exp(\|x_i\|^2)] < C$ for some constant $C < \infty$.
(v) $\gamma_i = \gamma_0 i^{-a}$ for some $1/2 < a < 1$.

Conditions (ii) imposes some moment conditions on the conditional density, and is satisfied if both $f_\varepsilon(\cdot|x_i)$ and $\frac{d}{d\varepsilon}f_\varepsilon(\cdot|x_i)$ are uniformly bounded along with a moment condition on x_i. Both conditions (ii) and (iii) are imposed here to establish the consistency of $\bar{\beta}_n$.

Condition (iii) can be viewed as a local identifiability condition, which ensures that the population loss function is locally strongly convex, and is reasonable as $\beta \mapsto q(\beta, Y_i)$ is convex.

Condition (iv) requires a tail condition on the regressors. It is stronger than the usual moment conditions for standard QR inference, but is required by both consistency and the asymptotic inference for S-subGD. In the literature, it is common to adopt stronger conditions on the regressors than on the error depending on the context. For example, Zheng et al. (2018) and Chen and Lee (2023) assumed the uniform boundedness of each of the covariates; the former paper emphasized that a linear quantile regression model with compact support for the regressors is most sensible to avoid the problem of quantile crossing.

One of the key technical steps in the proof of such one-observation-at-a-time updating, is to show that the nonlinear updating path is *linearizable*. Specifically, define $\Delta_i = \beta_i - \beta^*$, then the S-subGD updating can be rewritten using the estimation path:

$$\Delta_i = \Delta_{i-1} - \gamma_i \nabla Q\left(\beta_{i-1}\right) + \gamma_i \xi_i,$$

where $\xi_i = \nabla Q(\beta_{i-1}) - \nabla q(\beta_{i-1}, Y_i)$ is a martingale difference sequence. We then show that the path of Δ_i can be approximated by a linear path:

$$\Delta_i^1 := \Delta_{i-1}^1 - \gamma_i H \Delta_{i-1}^1 + \gamma_i \xi_i \quad \text{and} \quad \Delta_0^1 = \Delta_0,$$

where $H = \nabla^2 Q(\beta^*)$ is the population Hessian matrix. Such an *linearization*, as one of the technical steps in the proof, transfers the study of nonlinear SGD/S-subGD problems into linear problems.

Condition (v) gives a requirement for the learning rate, which is standard for *online learning*. Polyak and Juditsky (1992) showed that $a < 1$ is needed for the learning rate to decrease sufficiently slow so that the asymptotic normality of the Polyak–Ruppert averaging estimator holds. Meanwhile, for nonlinear models as in the QR model, it should not degenerate too slowly so $a > 1/2$ is also needed.

We extend Lee et al. (2022) (who assumed the sample loss function to be smooth and globally convex) to the quantile regression model, and establish a functional central limit theorem (FCLT) for the aggregated solution path of the S-subGD. Under Assumption 3.1, we establish the following FCLT:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{[nr]} \left(\beta_i - \beta^*\right) \Rightarrow \Upsilon^{1/2} W(r), \quad r \in [0,1], \tag{16}$$

where $\Rightarrow$ stands for the weak convergence in $\ell^\infty[0,1]$, $W(r)$ stands for a vector of the independent standard Wiener processes on $[0,1]$, and $\Upsilon := H^{-1} S H^{-1}$, with $S := \mathbb{E}[x_i x_i']\tau(1-\tau)$ and $H := \mathbb{E}[x_i x_i' f_\varepsilon(0|x_i)]$.

The FCLT in (16) states that the partial sum of the sequentially updated estimates β_i converges weakly to a rescaled Wiener process, with the scaling matrix equal to a square root of the asymptotic variance of the usual quantile regression estimator $\hat{\beta}_n$. Building on the FCLT, we propose a large-scale inference procedure.

For any $\ell \le d$ linear restrictions

$$H_0 : R\beta^* = c,$$

where R is an $(\ell \times d)$-dimensional known matrix of rank ℓ and c is an ℓ-dimensional known vector, the conventional Wald test based on $\hat{V}_n$ is asymptotically pivotal. We formally state the main theoretical result in the following theorem.

Theorem 3.1 (*Main Theorem*). *Suppose that $H_0 : R\beta^* = c$ holds with $\operatorname{rank}(R) = \ell$. Under Assumption 3.1, the FCLT in (16) holds and*

$$n\left(R\bar{\beta}_n - c\right)' \left(R\hat{V}_n R'\right)^{-1} \left(R\bar{\beta}_n - c\right) \xrightarrow{d} W(1)' \left(\int_0^1 \bar{W}(r)\bar{W}(r)' dr\right)^{-1} W(1),$$

where W is an ℓ-dimensional vector of the standard Wiener processes and $\bar{W}(r) := W(r) - rW(1)$.

As an important special case of Theorem 3.1, the t-statistic defined in (7) converges in distribution to the following pivotal limiting distribution: for each $j = 1, \ldots, d$,

$$\frac{\sqrt{n}\left(\bar{\beta}_{n,j} - \beta_j^*\right)}{\sqrt{\hat{V}_{n,jj}}} \xrightarrow{d} W_1(1) \left[\int_0^1 \left\{W_1(r) - rW_1(1)\right\}^2 dr\right]^{-1/2}, \tag{17}$$

where W_1 is a one-dimensional standard Wiener process.

3.1. Testing for homogeneity between two quantiles

Next, consider two quantiles $\tau_1, \tau_2 \in (0,1)$, and let $\beta_{\tau_1}^*$ and $\beta_{\tau_2}^*$ denote the corresponding coefficients at these two quantiles. We are interested in testing the heterogeneity at the two quantile levels:

$$H_0 : \beta_{\tau_1,sub}^* = \beta_{\tau_2,sub}^*.$$

which is to test whether particular subvectors of the coefficients at the two quantile levels are the same. For instance, the subvector could be the slope of a particular regressor of interest, and we are interested in testing whether its quantile effect stays the same when the quantile level changes from τ_1 to τ_2.

Both subvector coefficients can be iteratively estimated using S-subGD, whose stochastic paths $(\beta_i^1, \bar{\beta}_{i,sub}^1)$ and $(\beta_i^2, \bar{\beta}_{i,sub}^2)$ are constructed following the Algorithm (13)–(15). In particular, the stacked random scaling matrix $\bar{V}_{i,sub}$ corresponding to the two subvectors $(\bar{\beta}_{i,sub}^1, \bar{\beta}_{i,sub}^2)$ is updated as:

$$\begin{aligned} \theta_i &:= \begin{pmatrix} \bar{\beta}_{i,sub}^1 \\ \bar{\beta}_{i,sub}^2 \end{pmatrix}, \\ A_{i,sub} &= A_{i-1,sub} + i^2 \bar{\theta}_{i,sub}\bar{\theta}_{i,sub}', \quad b_{i,sub} = b_{i-1,sub} + i^2 \bar{\theta}_{i,sub}, \\ \bar{V}_i &= i^{-2}\left(A_i - \bar{\theta}_{i,sub} b_{i,sub}' - b_{i,sub}\bar{\theta}_{i,sub}' + \bar{\theta}_{i,sub}\bar{\theta}_{i,sub}' \sum_{s=1}^{i} s^2 \right). \end{aligned}$$

In this case, we let $G := (I, -I)$, and form a test statistic:

$$n(\bar{\beta}_{n,sub}^1 - \bar{\beta}_{n,sub}^2)'(G\bar{V}_{n,sub}G')^{-1}(\bar{\beta}_{n,sub}^1 - \bar{\beta}_{n,sub}^2).$$

We reject the null if the test statistic is large. The following theorem establishes the asymptotic null distribution.

Theorem 3.2. *Suppose that $H_0 : \beta_{\tau_1,sub}^* = \beta_{\tau_2,sub}^*$ holds. Under Assumption 3.1,*

$$n(\bar{\beta}_{n,sub}^1 - \bar{\beta}_{n,sub}^2)'(G\bar{V}_{n,sub}G')^{-1}(\bar{\beta}_{n,sub}^1 - \bar{\beta}_{n,sub}^2) \xrightarrow{d} W(1)'\left(\int_0^1 \bar{W}(r)\bar{W}(r)'dr\right)^{-1} W(1),$$

where W is an $2 \times \dim(\beta_{\tau_1,sub}^*)$-dimensional vector of the standard Wiener processes and $\bar{W}(r) := W(r) - rW(1)$.

4. Monte Carlo experiments

In this section we investigate the performance of the S-subGD random scaling method via Monte Carlo experiments. The main question of these numerical experiments is whether the proposed method is feasible for a large scale model.

The simulation is based on the following data generating process:

$$y_i = x_i'\beta^* + \varepsilon_i \quad \text{for} \quad i = 1, \dots, n,$$

where x_i is a $(d+1)$-dimensional covariate vector whose first element is 1 and the remaining d elements are generated from $\mathcal{N}(0, I_d)$. The error term ε_i is generated from $\mathcal{N}(0,1)$, and the parameter value β^* is set to be $(1,\dots,1)$. Without loss of generality, we focus on the second element, say $\beta_{(2)}^*$, of β^*. That is, we focus on subvector inference: the confidence interval for $\beta_{(2)}^*$ as well as testing the null hypothesis of $\beta_{(2)}^* = 1$. To accommodate a large scale model, the sample size varies in $n \in \{10^5, 10^6, 10^7\}$, and the dimension of x varies in $d \in \{10, 20, 40, 80, 160, 320\}$ in the first set of experiments. Later, we extend the dimension of x into $d \in \{500, 750, 1000, 1500\}$. The initial value β_0 is estimated by the convolution-type smoothed quantile regression (`conquer`) in He et al. (2023) and we do not burn in any observations. The learning rate is set to be $\gamma = \gamma_0 t^{-a}$ with $\gamma_0 = 1$ and $a = 0.501$. The simulation results are summarized from 1000 replications of each design.

We compare the performance of the proposed method with four additional alternatives:

(i) S-subGD: the proposed S-subGD random scaling method.
(ii) QR: the classical quantile regression method. Given the scale of models, we apply the Frisch–Newton algorithm after preprocessing and the Huber sandwich estimate under the local linearity assumption of the conditional density function by selecting ‘pfn’ and ‘nid’ options in R package `quantreg` (CRAN version 5.88).
(iii) CONQUER-plugin: estimates the parameter using the conquer method, and estimates the asymptotic variance by plugging in the parameter estimates; implemented using the R package `conquer` (CRAN version 1.3.0).
(iv) CONQUER-bootstrap: estimates the parameter using the conquer method, and applies the multiplier bootstrap method (see He et al. (2023) for details); implemented using the R package `conquer`. We set the number of bootstrap samples as 1000.
(v) SGD-bootstrap: estimates using the S-subGD method and conduct the inference by an online bootstrap method (see Fang et al. (2018) for details). We set the number of bootstrap samples as 1000.

We use the following performance measures: the computation time, the coverage rate, and the length of the 95% confidence interval. Note that the nominal coverage probability is 0.95. The computation time is measured by 10 replications on a separate desktop computer, equipped with an AMD Ryzen Threadripper 3970X CPU (3.7 GHz) and a 192 GB RAM. Other measures are computed by 1000 replications on the cluster system composed of several types of CPUs. We set the time budget as 10 h for a single estimation and the RAM budget as 192 GB.

Figs. 1–2 summarize the simulation result. To save space, we report the results of $d = 20, 80$, and 320. We provide tables for all the simulation designs in the appendix. First, we observe that S-subGD is easily scalable to a large simulation design. On the contrary, some methods cannot complete the computation within the time/memory budget. For instance, CONQUER-plugin does not work for any d when $n = 10^6$ or $n = 10^7$ because of the memory budget. QR shows a similar issue when $n = 10^7$ and $d = 320$. CONQUER-bootstrap cannot meet the 10-hour time budget for a single replication when $n = 10^7$ and $d = 320$. They are all denoted as ‘NA’ in the figures.

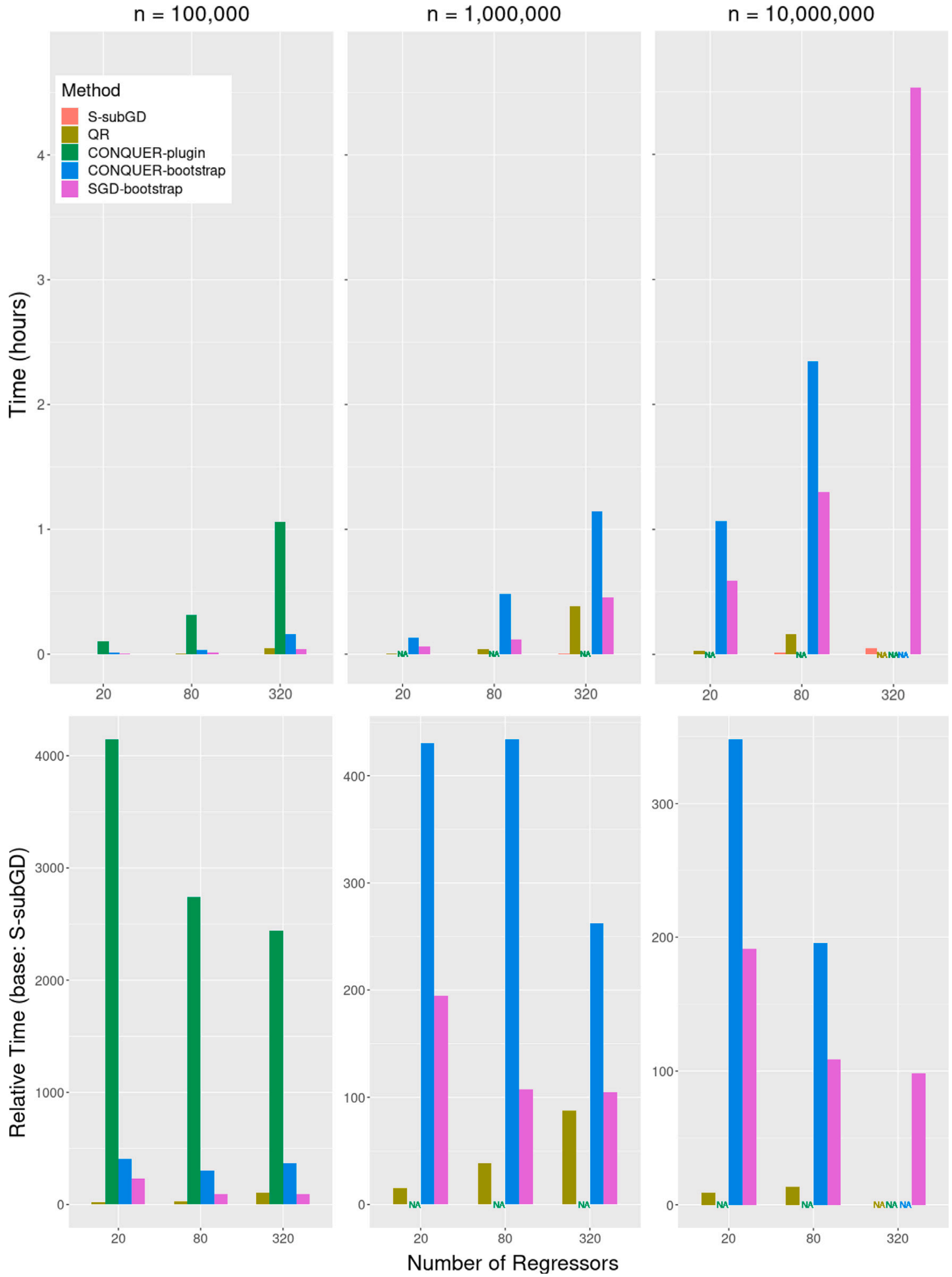

Fig. 1. Computation time.

Second, S-subGD outperforms the alternative in terms of computational efficiency. The bottom panel of Fig. 1 shows the relative computation time, and the relative speed improvement is by a factor of 10 or 100. In case of CONQUER-plugin, it is slower than

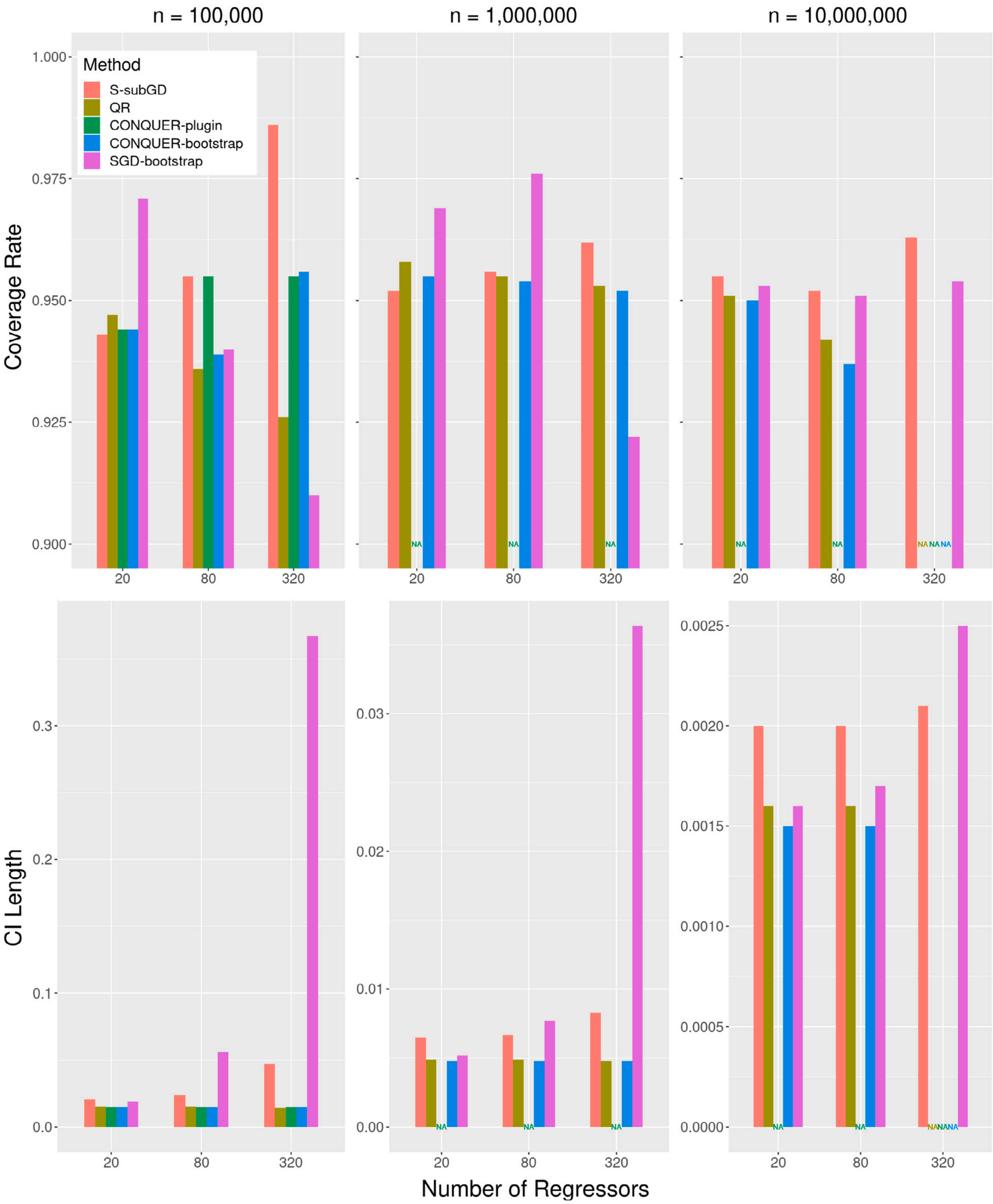

Fig. 2. Coverage and CI Length.

S-subGD in the scale of 1000's. If we convert them into the actual computation time, S-subGD takes 16 s for the design of $n = 10^6$ and $d = 320$. However, QR, CONQUER-bootstrap, and SGD-bootstrap take 1374 s (22 min 54 s), 4113 s (1 h 8 min 33 s), and 1643 s (27 min 23 s) on average of 10 replications.

Table 3
Performance of S-subGD: $n = 10^7$.

d	Time (s)	Coverage rate	CI length
10	4.80	0.965	0.0020
20	6.42	0.955	0.0020
40	11.02	0.954	0.0020
80	21.40	0.952	0.0020
160	29.87	0.953	0.0021
320	59.73	0.963	0.0021
500	154.12	0.945	0.0172
750	208.02	0.934	0.0359
1000	302.78	0.923	0.0461
1500	505.01	0.931	0.0645

Third, all methods shows satisfactory results in terms of the coverage rate and the length of the confidence interval. The top panel of Fig. 2 reports that coverage rates are all around 0.95 although S-subGD and SGD-bootstrap are slightly under-reject and over-reject when $d = 320$. The average lengths of the confidence interval are reported on the bottom panel. The performance of S-subGD is comparable to QR, CONQUER-plugin, and CONQUER-bootstrap. SGD-bootstrap shows much larger CI lengths, especially when $d = 320$.

In the second set of experiments, we stretch the computational burden by increasing the size of d up to $d = 1500$ when $n = 10^7$. Because of the challenging model scale, we use the cluster system for these exercises and check only the performance of S-subGD focusing on inference of a single parameter. Table 3 reports the simulation result along with different sizes of d. S-subGD still completes the computation within a reasonable time range. The average computation time is under 505 s (8 min 25 s) even for the most challenging design. The coverage rate and CI length becomes slightly worse than those of smaller d's but they still fall in a satisfactory range.

In sum, S-subGD performs well when we conduct a large scale inference problem in quantile regression. It is scalable to the range where the existing alternatives cannot complete the computation within a time/memory budget. In addition to computational efficiency, the coverage rate and CI lengths are also satisfactory. Thus, S-subGD provides a solution to those who hesitate to apply quantile regression with a large scale data for the computational burden.

5. Inference for the college wage premium

The study of the gender gap in the college wage premium has been a long-standing important question in labor economics. The literature has pointed out a stylized fact that the higher college wage premium for women as the major cause for attracting more women to attend and graduate from colleges than men (e.g., Goldin et al. (2006), Chiappori et al. (2009)). Meanwhile, Hubbard (2011) pointed out a bias issue associated with the "topcoded" wage data, where the use of the sample from Current Population Survey (CPS) often censors the wage data at a maximum value. As a remedy for it, Hubbard (2011) proposed to use quantile regression that is robust to censoring. Analyzing the CPS data during 1970–2008, he found no gender difference in the college wage premium later years once the topcoded bias has been accounted for.

We revisit this problem by estimating and comparing the college wage premium between women and men. While the data also contains the topcoded issue, the quantile regression is less sensitive to the values of upper-tail wages. By applying the new S-subGD inference to the ultra-big dataset we are using, we aim at achieving the following goals in the empirical study: (1) to identify (if any) the heterogeneous effects across quantiles; (2) to understand the trends in the college wage premium respectively for female and male; and (3) to understand the gender difference in the college wage premium.

We use the data from IPUMS USA (Ruggles et al., 2022) that contains several millions of workers. The main motivation of using ultra-large datasets for wage regressions is to deal with high-dimensional regressors within a simple parametric quantile regression model. As has been pointed out in the literature, work experience is an important factor in wage regressions but is difficult to be measured precisely. As career paths may be different across states and gender, one possible means of controlling for the experience is to flexibly interact workers' age, gender, and state fixed effects, which would create over one thousand regressors. Thus, a very large sample size would be desirable to obtain precise estimates. But the ultra-large dataset makes most existing inference procedures for quantile regression fail to work.

5.1. The data

We use the samples over six different years (1980, 1990, 2000–2015) from IPUMS USA at https://usa.ipums.org/usa/. In the years from 1980 to 2000, we use the 5% State sample which is a 1-in-20 national random sample of the population. In the remaining years, we use the American Community Survey (ACS) each year. The sampling ratio varies from 1-in-261 to 1-in-232 in 2001–2004, but it is set to a 1-in-100 national random sample of the population after 2005. To balance the sample size, we bunch the sample every 5 year after 2001. We also provide the estimation results using separate but smaller samples in 2005, 2010, and 2015 in the appendix, which is similar to those reported in this section.

We restrict our sample to $White$, $18 \leq Age \leq 65$, and $Wage \geq \$62$, which is a half of minimum wage earnings in 1980 ($\$3.10 \times 40$ h $\times 1/2$). $Wage$ denotes the implied weekly wage that is computed by dividing yearly earnings by weeks worked last

Table 4
Summary statistics.

Year	Sample size	$\mathbb{E}(Female)$	$\mathbb{E}(Educ)$	$\mathbb{E}(Educ\|Male)$	$\mathbb{E}(Educ\|Female)$
1980	3,659,684	0.390	0.433	0.444	0.416
1990	4,192,119	0.425	0.543	0.537	0.550
2000	4,479,724	0.439	0.600	0.578	0.629
2001–2005	2,493,787	0.447	0.642	0.619	0.670
2006–2010	4,708,119	0.447	0.663	0.631	0.701
2011–2015	4,542,874	0.447	0.686	0.646	0.735

Table 5
College wage premium: $\tau = 0.5$.

Year	Female	Male	Difference	Time (min)
$\tau = 0.5$				
1980	0.2552	0.1993	0.0559	7.6
	[0.2521,0.2583]	[0.1979,0.2007]	[0.0529,0.0589]	
1990	0.3700	0.2940	0.0761	6.7
	[0.3681,0.3720]	[0.2934,0.2946]	[0.0740,0.0781]	
2000	0.4124	0.3331	0.0792	7.1
	[0.4101,0.4147]	[0.3322,0.3341]	[0.0773,0.0812]	
2001–2005	0.4459	0.3732	0.0727	3.1
	[0.4435,0.4483]	[0.3708,0.3756]	[0.0688,0.0766]	
2006–2010	0.4782	0.4135	0.0648	7.6
	[0.4765,0.4799]	[0.4112,0.4157]	[0.0614,0.0681]	
2011–2015	0.4962	0.4383	0.0579	7.3
	[0.4940,0.4984]	[0.4337,0.4429]	[0.0544,0.0614]	

Notes. The male college premium is from $\hat{\beta}_2$ and the female college premium is from $\hat{\beta}_2+\hat{\beta}_3$. Thus, the college premium difference between male and female workers is from $\hat{\beta}_3$.

year. Since 2006, weeks worked last year are available only as an interval value and we use the midpoint of an interval. We only consider full-time workers who worked more than 30 h per week. Then, we compute the *real* wage using the personal consumption expenditures price index (PCEPI) normalized in 1980. The data cleaning leaves us with 3.6-4.7 million observations besides 2001–2005, where we have around 2.5 million observations. Table 4 reports some summary statistics on the key variables, where $Educ$ denotes an education dummy for some college or above. The table confirms that female college education has increased substantially over the years and female workers received more college education than male workers since 1990.

5.2. The quantile regression model

We use the following baseline model:

$$\log(Wage_i) = \beta_0 + \beta_1 Female_i + \beta_2 Educ_i + \beta_3 Female_i \cdot Educ_i + \theta_1' X_i + \theta_2'(X_i \cdot Female_i) + \varepsilon_i,$$

where $Wage_i$ is a real weekly wage in 1980 terms, $Female_i$ is a female dummy, $Educ_i$ is an education dummy for some college or above, and X_i is a vector of additional control variables. For control variable X_i, we use 12 age group dummies with a four-year interval, 51 states dummies (including D.C.), and their interactions. Note that $(X_i \cdot Female_i)$ implies that there exist up to 3-way interactions. The model contains 1226 covariates in total. We also add 4 additional year dummies for the 5-year combined samples after 2001. We estimate the model using the proposed S-subGD method over 9 different quantiles: $\tau = 0.1, 0.2, \dots, 0.9$. We obtain the starting value of S-subGD inference by applying `conquer` to the 10% subsample.[5] For the learning rate $\gamma_0 t^{-a}$, we set $a = 0.501$ and γ_0 by the rule-of-thumb approach described in Section 2.4. As an sensitivity analysis, we also vary a over 17 equi-spaced points in $[0.501, 0.661]$. The results are quite robust to the choice of a as reported in the online appendix. We use the Compute Canada cluster system equipped with the AMD Milan CPUs whose clock speed is 2.65 GHz and with 650 GB of RAMs. For comparison, we also try to estimate the model using the standard QR method with the 'quantreg' R package, but it fails to compute the result in the given cluster environments.

5.3. The results

Figs. 3–4 and Table 5 summarize the estimation results. We also provide the full estimation results in the appendix.

Our results share several interesting features about the estimated college wage premium. First, some degrees of heterogeneity are found over different quantiles. At the upper tail quantile, $\tau = 0.9$, male college premium is slightly higher than the female one

[5] Recall that X_i contains a large number of dummy variables in this empirical application. If the subsample size is too small, `conquer` may face a singularity issue and cannot compute the initial value. Alternatively, one may use a randomized algorithm of Yang et al. (2013).

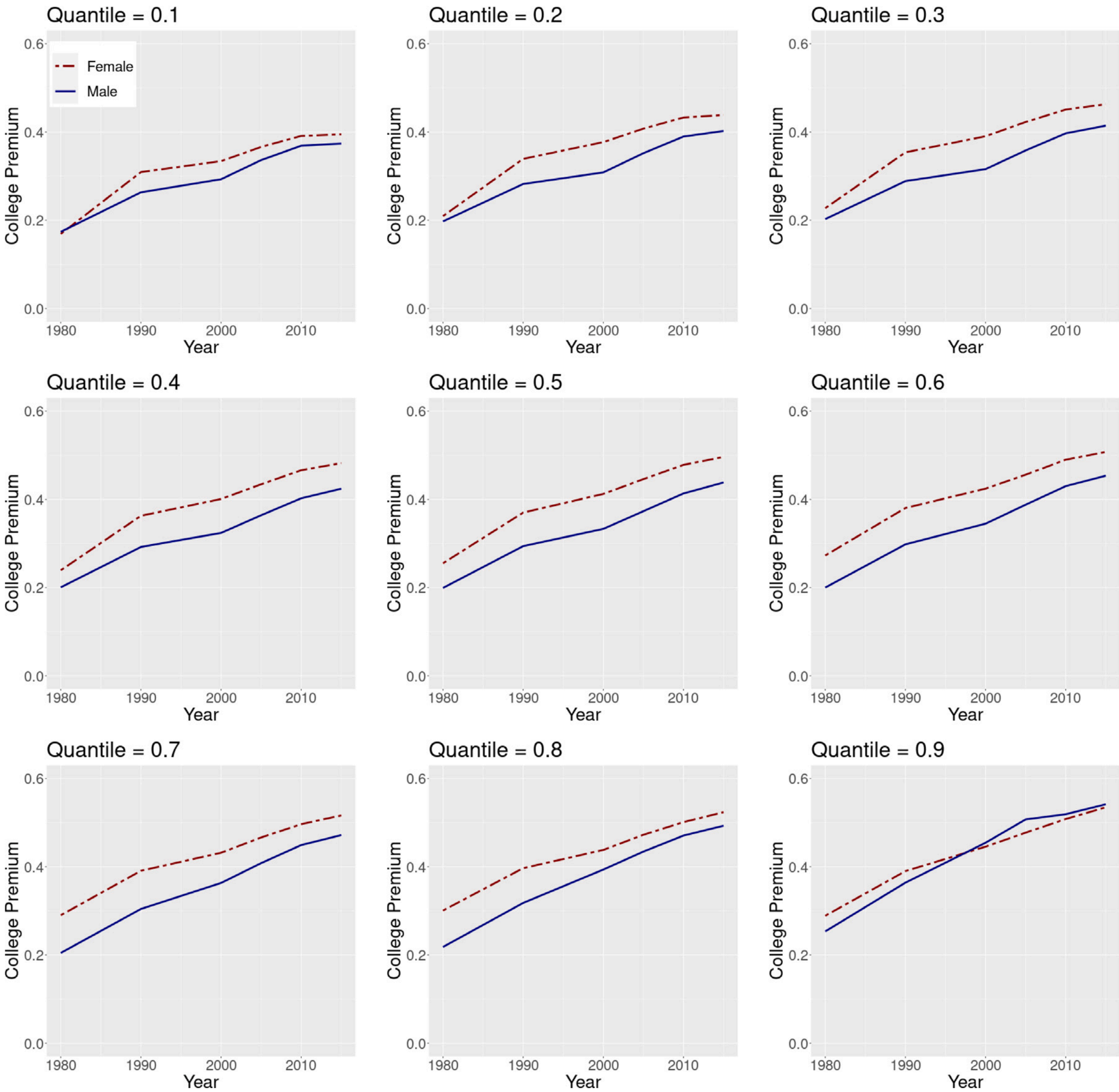

Fig. 3. College Wage Premium: Combining 5-Year Data.

since 2000, while female college premiums are higher over all years in other quantiles. This shows an interesting feature about high-income individuals that cannot be viewed by mean regression, though sensible economic interpretations might be subject to debates. Also, note that the 95% confidence intervals are slightly wider for tail quantiles ($\tau = 0.1$ and 0.9, respectively).

Second, college premiums increase over time for both male and female. This result coincides with the overall findings in the literature. It is interesting that college premiums get flatter since 2010 for lower quantiles ($\tau \leq 0.5$).

Third, when we focus on the median ($\tau = 0.5$) as reported in Table 5, we observe that the female-male college premium difference shows an inverse "U-shape" pattern. In addition, the difference is always significantly positive, which is different from the result in Hubbard (2011). He found the gender difference to be insignificant starting from around the year 2000 from the CPS data. We also note that the computation time is within a reasonable range meanwhile it is not feasible to estimate the model with the alternative approaches.

6. Discussions

We have proposed an inference method for large-scale quantile regression to analyze datasets whose sizes are of order $(n, p) \sim (10^7, 10^3)$. Based on the stochastic subgradient descent updates, our method runs fast and constructs asymptotically pivotal statistics via random scaling.

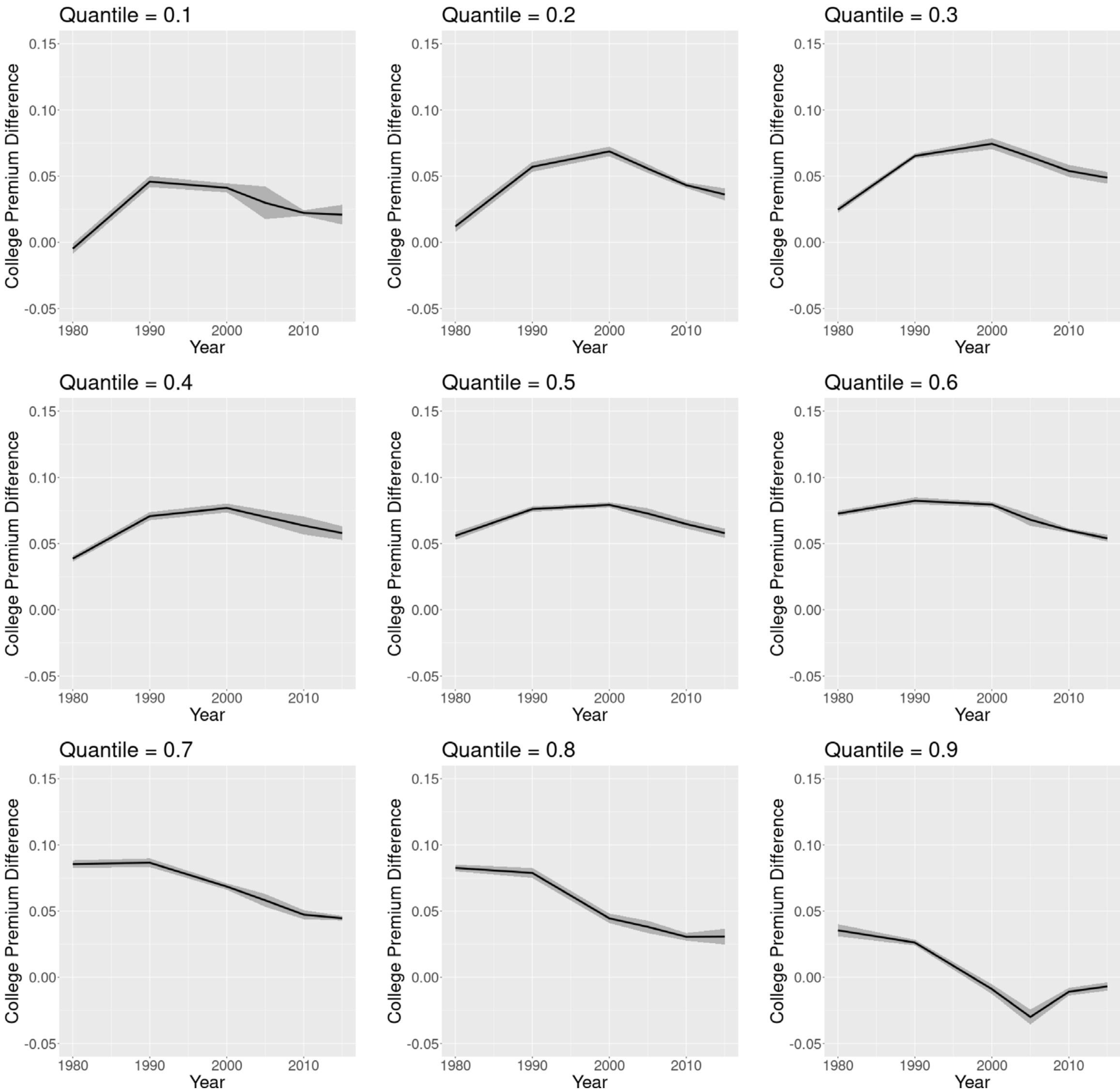

Fig. 4. Difference of College Premium: Combining 5-Year Data.

As an important extension, one can embed variable selections in the S-subGD framework. At the i th update, define the t-statistic $t_{i,j} = \bar{\beta}_{i,j}/\sqrt{V_{i,jj}}$. for $j = 1, \dots, d$. Then, variable j is selected if

$$|t_{i,j}| > \lambda$$

for a tuning parameter λ. For instance, we can set $\lambda = 6.747$ which is the critical value for the 95% confidence level of the asymptotic distribution. For each variable j, we obtain a cumulative selection path along the iteration:

$$SP(j,i) = \frac{1}{i}\sum_{k=1}^{i} 1\{|t_{k,j}| > \lambda\}.$$

The SP path provides rich information about variable selections, which is particularly attractive when the signal-noise ratio is relatively weak. We would like to leave the theoretical justification of the selection path for future studies.

Besides the variable selection, there are a few more extensions worth pursuing in future research. First, we may build on Chernozhukov et al. (2022) to develop fast inference for the quantile regression process. Second, while we focus on the regular quantile regression models where the quantile is assumed to be bounded away from both zero and one, our framework is also potentially applicable to extreme quantile regression models (see, e.g., Chernozhukov et al., 2016, for a review). The ability of handling ultra-large datasets is also appealing for extreme quantile regression models because the resulting sample size is considerably larger at the extreme quantiles. A formal treatment for these cases is out of the scope of this paper, and we leave it for future studies. Third, Shan and Yang (2009) considered, among other things, an online setting for which time series data arrive sequentially and several quantile estimators are combined with weights being updated sequentially. It is an interesting topic to study the properties of our proposed method with time series data. Finally, Liu et al. (2023) developed online inference on population quantiles with privacy concerns. It is another research topic to adapt our inference method for privacy-focused applications.

Appendix A. Proofs

In the appendix, we provide the proof of Theorem 3.1. Especially, we prove Theorem A.1 which is a more general version than Theorem 3.1. Theorem A.1 includes quantile regression as a special case but can be of independent interest. To do so, we first present high-level regularity conditions, which we will verify for quantile regression in Theorem A.2.

A.1. General theory for M-estimation that allows possibly non-smooth and locally-strongly-convex loss

We formalize a general inferential theory for M-estimations that allows non-smooth (like quantile regression) and non-globally-strongly-convex loss functions (but locally strongly convex), in the online fashion.

Recall that β^* is characterized by

$$\beta^* := \arg\min_{\beta\in\mathbb{R}^d} Q(\beta),$$

where $Q(\beta) := \mathbb{E}[q(\beta, Y_i)]$ is the population loss function. In this subsection, it can be the loss function for general M-estimations, not just the quantile regression. The updating rule is

$$\beta_i = \beta_{i-1} - \gamma_i \nabla q\left(\beta_{i-1}, Y_i\right),$$

where ∇q belongs to the subgradient of q. Next, define

$$\xi_i(\beta) := \nabla Q(\beta) - \nabla q\left(\beta, Y_i\right),$$

and $\xi_i := \xi_i(\beta_{i-1})$.

We impose the following conditions.

Assumption A.1 (*IID Data*)**.** Data $\{Y_i : i = 1, \dots, n\}$ are i.i.d.

Assumption A.1 restricts our attention to i.i.d. data. Assumption A.1 implies among other things that the sequence $\{\xi_i\}_{i\geq 1}$ is a martingale-difference sequence (mds) defined on a probability space $(\Omega, \mathcal{F}, \mathcal{F}_i, P)$. That is, $\mathbb{E}(\xi_i|\mathcal{F}_{i-1}) = 0$ almost surely.

Assumption A.2 (*Population Criterion*)**.** The real-valued function $Q(\beta) := \mathbb{E}\left[q\left(\beta, Y_i\right)\right]$ is twice continuously differentiable, with:

(i) $\sup_{\beta\in\Theta} \|\nabla^2 Q(\beta)\| < \infty$.

(ii) There is $K_1 > 0$ and $\epsilon > 0$ that for all $\|\beta - \beta^*\| \leq \epsilon$,

$$\|\nabla^2 Q(\beta) - \nabla^2 Q(\beta^*)\| \leq K_1 \|\beta - \beta^*\|.$$

(iii) In addition, $\nabla^2 Q(\beta)$ is positive semi-definite for all β.

Assumption A.2 imposes mild regularity conditions on the population criterion function $Q(\beta)$. It is similar to Assumption 3.3 of Polyak and Juditsky (1992), but with an important extension that allows for non-differentiable loss functions as in quantile regression.

Assumption A.3 (*Learning Rate*)**.** $\gamma_i = \gamma_0 i^{-a}$ for some $1/2 < a < 1$.

Assumption A.3 is the standard condition on the learning rate.

Assumption A.4 (*Local Strong Convexity*)**.** There are constants $\epsilon > 0$ and $c_0 > 0$ that satisfy

$$\inf_{\|\beta-\beta^*\|<\epsilon} \lambda_{\min}\left(\nabla^2 Q(\beta)\right) > c_0.$$

The key difference between Assumption A.4 and those of Polyak and Juditsky (1992) is that in our setting, the strong convexity is imposed only locally on a neighborhood of the true value. In contrast, Polyak and Juditsky (1992) imposed the global strong convexity. They also imposed additional global conditions, which are relaxed here. This relaxation is important for the quantile regression model, which may not be globally strongly convex.

Assumption A.5 (*Sufficient Conditions for Consistency*)**.** The sequence $\left\{\xi_i\right\}_{i\geq 1}$ satisfy the following additional conditions.

(i) $\mathbb{E}\left[\|\xi_i\|^6 \exp\left(\left(1 + \|\xi_i\|^4\right)^{1/2}\right) \Big| \mathcal{F}_{i-1}\right] < \infty \quad \text{a.s.}$

(ii) $S_i(\beta) := \mathbb{E}\left[\xi_i(\beta)\xi_i(\beta)'|\mathcal{F}_{i-1}\right]$ is uniformly Lipschitz continuous, that is,

$$\left\|S_i\left(\beta_1\right) - S_i\left(\beta_2\right)\right\| \leq L\|\beta_1 - \beta_2\| \text{ a.s. for some } L < \infty.$$

(iii) In addition, for some $p \geq (1-a)^{-1}$, $\mathbb{E}\left\|\xi_i\right\|^{2p}$ is bounded. Here, a is defined in Assumption A.3.

Conditions (i) and (ii) in Assumption A.5 are imposed to apply Theorem 2 of Gadat and Panloup (2023). In addition, under Assumption A.5(i)-(ii) combined with the local strong convexity in Assumption A.4, a global Kurdyka-Łojasiewicz condition in Gadat and Panloup (2023) is satisfied. From here, we can establish the consistency. Condition (iii) of Assumption A.5 is an additional moment condition to strengthen the results for the FCLT.

Assumption A.6 (*Stochastic Disturbances*).

(i) For $K_2 > 0$ and for all $i \geq 1$,

$$\|\nabla Q(\beta_{i-1})\|^2 + \mathbb{E}(\|\nabla q(\beta_{i-1}, Y_i)\|^2|\mathcal{F}_{i-1}) \leq K_2(1 + \|\beta_{i-1}\|^2) \ \ a.s.$$

(ii) Denote $\nabla q_i^* := \nabla q\left(\beta^*, Y_i\right)$. Then

(a) There is a symmetric and positive definite matrix S, such that as $i \to \infty$,

$$\mathrm{Var}\left(\nabla q_i^*|\mathcal{F}_{i-1}\right) = S_i(\beta^*) \xrightarrow{P} S,$$

where $S_i(\cdot)$ is as defined in Assumption A.5.

(b) As $C \to \infty$,

$$\sup_i \mathbb{E}[\|\nabla q_i^*\|^2 I(\|\nabla q_i^*\| > C|\mathcal{F}_{i-1})] \xrightarrow{P} 0.$$

(iii) There is a function $h(\cdot)$ such that $h(x) \to 0$ as $\|x\| \to 0$, which satisfies: for all i large enough, almost surely,

$$\mathbb{E}(\|\nabla q\left(\beta_{i-1}, Y_i\right) - \nabla q\left(\beta^*, Y_i\right)\|^2|\mathcal{F}_{i-1}) \leq h(\beta_{i-1} - \beta^*).$$

We now establish the general theorem under these assumptions.

Theorem A.1 (*General Theorem*). *Under Assumptions A.1–A.6,*

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{[nr]}\left(\beta_i - \beta^*\right) \Rightarrow \Upsilon^{1/2}W_d(r), \quad r \in [0,1], \tag{18}$$

where W_d is a d-dimensional vector of the standard Wiener processes; $\Rightarrow$ stands for the weak convergence in $\ell^\infty[0,1]$, and $\Upsilon := H^{-1}SH^{-1}$, with S being the probability limit of Var $\left(\nabla q\left(\beta^*, Y_i\right)|\mathcal{F}_{i-1}\right)$ and $H := \nabla^2 Q(\beta^*)$.

In addition, under $H_0 : R\beta^* = c$ for an $\ell \times d$ matrix R with $\mathrm{rank}(R) = \ell$,

$$n\left(R\bar{\beta}_n - c\right)'\left(R\widehat{V}_n R'\right)^{-1}\left(R\bar{\beta}_n - c\right) \xrightarrow{d} W_\ell(1)'\left(\int_0^1 \bar{W}_\ell(r)\bar{W}_\ell(r)'dr\right)^{-1} W_\ell(1),$$

where $\bar{W}_\ell(r) := W_\ell(r) - rW_\ell(1)$.

A.2. Proof of Theorem A.1

Proof of Theorem A.1.

We extend the proof of Polyak and Juditsky (1992) and Lee et al. (2022) in two aspects: allow non-globally-strong-convex and possibly nonsmooth loss functions. For the former, we first establish the consistency. Then we can focus on a local neighborhood of the true value. It is under Assumption A.4 that the loss function is locally strongly convex. The formal proof is divided in the following steps. Some of them are delegated to Lemma A.1 – A.3 in the online supplement.

step 1: consistency. Lemma A.1 shows that

$$\bar{\beta}_n \to^P \beta^*.$$

step 2: local-linearization. Rewrite (4) as

$$\beta_i = \beta_{i-1} - \gamma_i \nabla Q\left(\beta_{i-1}\right) + \gamma_i \xi_i. \tag{19}$$

Let $\Delta_i := \beta_i - \beta^*$ and $\bar{\Delta}_i := \bar{\beta}_i - \beta^*$ to denote the errors in the ith iterate and that in the average estimate at i, respectively. Then, subtracting β^* from both sides of (19) yields that

$$\Delta_i = \Delta_{i-1} - \gamma_i \nabla Q\left(\beta_{i-1}\right) + \gamma_i \xi_i.$$

Now consider the following linear process Δ_i^1, defined as:

$$\Delta_i^1 := \Delta_{i-1}^1 - \gamma_i H \Delta_{i-1}^1 + \gamma_i \xi_i \quad \text{and} \quad \Delta_0^1 = \Delta_0,$$

where $H = \nabla^2 Q(\beta^*)$. Furthermore, for $r \in [0,1]$, introduce a partial sum process

$$\bar{\Delta}_i(r) := i^{-1} \sum_{j=1}^{[ir]} \Delta_j, \quad \bar{\Delta}_i^1(r) := i^{-1} \sum_{j=1}^{[ir]} \Delta_j^1.$$

This step establishes a uniform approximation of the partial sum process $\Delta_i^r(r)$ to $\bar{\Delta}_i^1(r)$. To this end, we adopt Part 4 in the proof of Polyak and Juditsky (1992)'s Theorem 2. To adopt their proof, we need to establish that the local convexity of the loss function can ensure that the nonlinear updating process Δ_i can be approximated locally (as $\beta_i \to^P \beta^*$) by the linear process Δ_i^1. Indeed, as Lemma A.2 shows, in the neighborhood of β^*, this is the case. In fact, Lemma A.2 verifies that the required conditions, Assumptions 3.1 and 3.2 of Polyak and Juditsky (1992), hold locally. In addition, carefully examining their proof of Part 4, it is only required that the population loss function be twice differentiable, as the linear process Δ_i^1 only depends on the Hessian of the population loss function.

Therefore, with Lemma A.2, the proof of Part 4 in Polyak and Juditsky (1992) goes through, which establishes:

$$\sqrt{i} \sup_r \left\| \bar{\Delta}_i(r) - \bar{\Delta}_i^1(r) \right\| = o_p(1).$$

Hence it suffices to analyze the sequence $\bar{\Delta}_i^1$ in place of $\bar{\Delta}_i$, and establish the weak convergence of $\bar{\Delta}_i^1$.

step 3: weak convergence of $\sqrt{i}\bar{\Delta}_i^1(r)$. Following the decomposition in (A10) in Polyak and Juditsky (1992), write

$$\sqrt{i}\bar{\Delta}_i^1(r) = I^{(1)}(r) + I^{(2)}(r) + I^{(3)}(r),$$

where

$$I^{(1)}(r) := \frac{1}{\gamma_0\sqrt{i}} \alpha_{[ir]} \Delta_0, \; I^{(2)}(r) := \frac{1}{\sqrt{i}} \sum_{j=1}^{[ir]} H^{-1} \xi_j, \text{ and } I^{(3)}(r) := \frac{1}{\sqrt{i}} \sum_{j=1}^{[ir]} w_j^{[ir]} \xi_j,$$

where $\alpha_i \le K$ and $\left\{ w_j^{[ir]} \right\}$ is a bounded sequence such that $i^{-1} \sum_{j=1}^{i} \left\| w_j^i \right\| \to 0$. Then, $\sup_r \left\| I^{(1)}(r) \right\| = o_p(1)$. Suppose for now that $\mathbb{E} \sup_r \| I^{(3)} \|^p = o(1)$ for some $p \ge 1$. Bounding $I^{(3)}$ requires some involved arguments, as $w_j^{[ir]} \xi_j$ is not mds, even though ξ_j is (note that $w_j^{[ir]} \xi_j$ is indexed by r). The next step of the proof develops new techniques to bound this term.

We now apply the FCLT for mds, see e.g. Theorem 4.2 in Hall and Heyde (1980). To verify its sufficient conditions, we prove Lemma A.3. Then Lemma A.3 is sufficient to verify the conditions in Hall and Heyde (1980) using the same argument of Part 1 of the proof in Polyak and Juditsky (1992). Hence, $I^{(2)}$ converges weakly to a rescaled Wiener process $\Upsilon^{1/2} W(r)$. Hence

$$\sqrt{i}\bar{\Delta}_i^1(r) \Rightarrow \Upsilon^{1/2} W(r).$$

step 4: uniform convergence of $I^{(3)}(r)$. Let $S_i = \sum_{j=1}^{i} w_j^i \xi_j$. Let $p > (1-a)^{-1}$ and note that

$$\mathbb{E} \sup_r \left\| I^{(3)}(r) \right\|^{2p} \le t^{-p} \mathbb{E} \sup_r \left\| S_{[ir]} \right\|^{2p} \le t^{-p} \sum_{m=1}^{i} \mathbb{E} \left\| S_m \right\|^{2p}.$$

Note that for each m, S_m is the sum of mds. Hence, due to Burkholder's inequality (e.g., Theorem 2.10 in Hall and Heyde, 1980),

$$\begin{aligned} \mathbb{E} \left\| S_m \right\|^{2p} &\le C_p \mathbb{E} \left| \sum_{j=1}^{m} \left\| w_j^m \right\|^2 \left\| \xi_j \right\|^2 \right|^p \\ &= C_p \sum_{j_1,\dots,j_p=1}^{m} \left\| w_{j_1}^m \right\|^2 \cdots \left\| w_{j_p}^m \right\|^2 \mathbb{E} \left\| \xi_{j_1} \right\|^2 \cdots \left\| \xi_{j_p} \right\|^2, \end{aligned}$$

where the universal constant C_p depends only on p. Note that $\mathbb{E} \left\| \xi_{j_1} \right\|^2 \cdots \left\| \xi_{j_p} \right\|^2$ is bounded since $\mathbb{E} \left\| \xi_j \right\|^{2p}$ is bounded. Also, the boundedness of $\left\| w_j^m \right\|$ yields $\sum_{j=1}^{m} \left\| w_j^m \right\|^b = O\left(\sum_{j=1}^{m} \left\| w_j^m \right\| \right)$ for any b. By Lemma 2 in Zhu and Dong (2021), $\sum_{j=1}^{m} \left\| w_j^m \right\| = o(m^a)$. These facts yield that

$$\mathbb{E} \left\| S_m \right\|^{2p} = O\left(\left(\sum_{j=1}^{m} \left\| w_j^m \right\| \right)^p \right) = o(m^{ap}), \tag{20}$$

which holds uniformly for m, k. It in turn implies the desired result that

$$\mathbb{E} \sup_r \left\| I^{(3)}(r) \right\|^{2p} \le t^{-p} \sum_{m=1}^{i} o(m^{ap}) = o\left(t^{1+ap-p} \right) = o(1).$$

step 5: FCLT. By the previous three steps:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{[nr]} (\beta_i - \beta^*) = \sqrt{n}\bar{\Delta}_n(r) = \sqrt{n}\bar{\Delta}_n^1(r) + o_P(1) = \frac{1}{\sqrt{n}} \sum_{j=1}^{[nr]} H^{-1} \xi_j + o_P(1). \tag{21}$$

We have $\frac{1}{\sqrt{n}}\sum_{j=1}^{[nr]} H^{-1}\xi_j \Rightarrow \Upsilon^{1/2}W(r)$. This establishes the FCLT in (16).

step 6: random scaling. Now let $C_n(r) := R\frac{1}{\sqrt{n}}\sum_{i=1}^{[nr]}\left(\beta_i - \beta^*\right)$. Also let $\Lambda = (R\Upsilon R')^{1/2}$. It is well-defined and invertible when $l \le d$. By the FCLT in the previous step, for some vector of independent standard Wiener process $W^*(r)$,

$$C_n(r) \Rightarrow \Lambda W^*(r).$$

In addition, $R\widehat{V}_n R' = \frac{1}{n}\sum_{s=1}^{n}[C_n(\frac{s}{n}) - \frac{s}{n}C_n(1)][C_n(\frac{s}{n}) - \frac{s}{n}C_n(1)]'$. Here the sum is also an integral over r as $C_n(r)$ is a partial sum process, and $R(\bar\beta_n - \beta^*) = \frac{1}{\sqrt{n}}C_n(1)$. Hence $n\left(R\bar\beta_n - c\right)'\left(R\widehat{V}_n R'\right)^{-1}\left(R\bar\beta_n - c\right)$ is a continuous functional of $C_n(\cdot)$. Then the continuous mapping theorem proves the theorem. □

A.3. Proof of Theorem 3.1

Proof of Theorem 3.1. This follows directly from Theorem A.1 because all the regularity conditions in Theorem A.1 are verified in Theorem A.2. □

We verify the high-level assumptions with more primitive conditions for the quantile regression model.

Theorem A.2 (*Verifying High-Level Conditions for Quantile Regression*). *Suppose that Assumption 3.1 holds. Then, Assumptions A.1–A.6 hold with $S = \mathbb{E}[x_i x_i']\tau(1-\tau)$ and $H = \mathbb{E}[x_i x_i' f_\varepsilon(0|x_i)]$.*

Proof of Theorem A.2.

Verify Assumption A.1. It is directly imposed.

Verify Assumption A.2. It is straightforward to verify the twice differentiability, where

$$\nabla Q(\beta) = \mathbb{E}x_i[P(\varepsilon_i \le x_i'(\beta - \beta^*)|x_i) - \tau].$$

and

$$\nabla^2 Q(\beta) = G(\beta) = \mathbb{E}[x_i x_i' f_\varepsilon(x_i'(\beta - \beta^*)|x_i)].$$

(i) $\sup_{\beta\in\Theta}\|\nabla^2 Q(\beta)\| \le \sup_{b\in\Theta}\|\mathbb{E}[x_i x_i' f_\varepsilon(x_i' b)|x_i]\| < \infty$.

(ii) For all $\|\beta - \beta^*\| \le \varepsilon$,

$$\|\nabla^2 Q(\beta) - \nabla^2 Q(\beta^*)\| \le \|\nabla^3 Q(\widetilde\beta)\|\|\beta - \beta^*\|$$

for some $\widetilde\beta$. Here $\nabla^3 Q(\beta)$ is a $\dim(\beta)\times\dim(\beta)^2$ matrix, whose j th row is given by $[\text{vec}\nabla^2(\partial_j Q(\beta))]'$. We note that

$$\|\nabla^3 Q(\widetilde\beta)\| \le \sup_{|\beta-\beta^*|<\epsilon}\mathbb{E}\|x_i\|^3\left|\frac{d}{d\varepsilon}f_\varepsilon(x_i'(\beta - \beta^*)|x_i)\right| < K_1.$$

The last inequality follows from Assumption 3.1(ii).

(iii) It is clear that $\mathbb{E}[x_i x_i' f_\varepsilon(x_i'(\beta - \beta^*)|x_i)]$ is globally semipositive definite.

Verify Assumption A.3. It is directly imposed.

Verify Assumption A.4. This is imposed by Assumption 3.1(i) that the eigenvalues of $\mathbb{E}[x_i x_i' f_\varepsilon(x_i'(\beta - \beta^*)|x_i)]$ are locally bounded away from zero.

Verify Assumption A.5. For (i), we have for any $r > 0$, $\|\xi_i\|^r \le C(\|x_i\|^r + \mathbb{E}\|x_i\|^r)$. Hence

$$\mathbb{E}\left[\|\xi_i\|^6\exp\left(\left(1 + \|\xi_i\|^4\right)^{1/2}\right)|\mathcal{F}_{i-1}\right] \le C\mathbb{E}[\|x_i\|^6 + 1)\exp(\|x_i\|^2|\mathcal{F}_{i-1}] < C.$$

For (ii), it is a straightforward application of the Cauchy–Schwarz and triangular inequalities. For instance, $\mathbb{E}\|x_i\|^2|I\{\varepsilon_i < \beta_1\} - I\{\varepsilon_i < \beta_2\}| \le \sup_b \mathbb{E}\|x_i\|^2 f_\varepsilon(x_i' b)\|\beta_1 - \beta_2\|$. We omit the details. For (iii), note, for all β, $\mathbb{E}\|\xi_i(\beta)\|^{2p} \le 2\times 4^p\mathbb{E}\|x_i\|^{2p} < \infty$.

Verify Assumption A.6. (i) Write

$$\|\nabla Q\left(\beta_{i-1}\right)\|^2 \le 4(\mathbb{E}\|x_i\|)^2 \le K$$
$$\mathbb{E}(\|\nabla q(\beta_{i-1}, Y_i)|^2\|\mathcal{F}_{i-1}) \le 4\mathbb{E}(\|x_i\|^2|\mathcal{F}_{i-1}) \le 4\mathbb{E}\|x_i\|^2 \le K.$$

This implies

$$\|\nabla Q\left(\beta_{i-1}\right)\|^2 + \mathbb{E}(\|\nabla q(\beta_{i-1}, Y_i)\|^2|\mathcal{F}_{i-1}) \le 2K \le 2K(1 + \|\beta_{i-1}\|^2)\ \ a.s.$$

(ii) Here $\nabla q_i^* = \xi_i(\beta^*) = x_i[I\{\varepsilon_i < 0\} - \tau]$. Clearly, $\mathbb{E}(\xi_i(\beta^*)|\mathcal{F}_{i-1}) = 0$ and (a) $\text{Var}(\xi_i(\beta^*)|\mathcal{F}_{i-1}) = \mathbb{E}[x_i x_i']\tau(1-\tau) = S$. (b) Also, as $C \to \infty$,

$$\sup_i \mathbb{E}[\|\xi_i(\beta^*)\|^2 I(\|\xi_i(\beta^*)| > C|\mathcal{F}_{i-1})] \le \sup_i(\mathbb{E}\|\xi_i(\beta^*)\|^3|\mathcal{F}_{i-1})^{2/3}P(\|\xi_i(\beta^*)\| > C|\mathcal{F}_{i-1})^{1/3}$$
$$\le 4(\mathbb{E}\|x_i\|^3)^{2/3}\mathbb{E}(\|\xi_i(\beta^*)\||\mathcal{F}_{i-1})^{1/3}C^{-1/3}\xrightarrow{P} 0.$$

(iii) Recall $\nabla q(\beta, Y_i) := x_i[I\{y_i \le x_i'\beta\} - \tau]$. Write

$$A_i = x_i[I\{\varepsilon_i < x_i'(\beta_{i-1} - \beta^*)\} - I\{\varepsilon_i < 0\}].$$

Hence for $h(x) = 2\sup_b \mathbb{E}\|x_i\|^3 f_\varepsilon(x_i'b|x_i)\|x\|$,

$$\mathbb{E}(\|\nabla q(\beta_{i-1}, Y_i) - \nabla q(\beta^*, Y_i)\|^2|\mathcal{F}_{i-1}) \le \mathbb{E}(\|A_i\|^2|\mathcal{F}_{i-1}) \le h(\beta_{i-1} - \beta^*). \quad \square$$

A.4. Proof of Theorem 3.2

Proof of Theorem 3.2.

By (21), uniformly in r, and $d = 1, 2$,

$$i^{-1/2}\sum_{j=1}^{[ir]}[\beta_{j,sub}^d - \beta_{\tau_d,sub}^*] = \frac{1}{\sqrt{i}}\sum_{j=1}^{[ir]}(H_d^{-1})_{sub}\xi_{j,d} + o_P(1)$$

where $(H_d^{-1})_{sub}$ denotes the rows of $H_d = \mathbb{E}[x_i x_i' f_{\varepsilon,d}(0|x_i)]$ corresponding to the subvector $\beta_{\tau_d,sub}^*$; $f_{\varepsilon,d}(|x)$ denotes the conditional density of $\varepsilon_{i,d}$ for $d = 1, 2$,

$$\xi_{i,d}(\beta) := \mathbb{E}x_i[P(\varepsilon_{i,d} \le x_i'(\beta - \beta_{\tau_d}^*)|x_i) - \tau_d] - x_i[I\{\varepsilon_{i,d} \le x_i'(\beta - \beta_{\tau_d}^*)\} - \tau_d].$$

and $\xi_{i,d} := \xi_{i,d}(\beta_{i-1}^d)$. Then we have

$$\sqrt{i}\bar{\Delta}_i(r) := i^{-1/2}\sum_{j=1}^{[ir]}\begin{pmatrix}\beta_{j,sub}^1 - \beta_{\tau_1,sub}^*\\ \beta_{j,sub}^2 - \beta_{\tau_2,sub}^*\end{pmatrix} = \frac{1}{\sqrt{i}}\sum_{j=1}^{[ir]}\bar{H}u_j + o_P(1), \quad \bar{H} = \begin{pmatrix}(H_1^{-1})_{sub} & 0\\ 0 & (H_2^{-1})_{sub}\end{pmatrix}$$

$$u_j := u_j(\beta_{j-1}^1, \beta_{j-1}^2), \quad u_j(\beta^1, \beta^2) := \begin{pmatrix}\xi_{j,1}(\beta^1)\\ \xi_{j,2}(\beta^2)\end{pmatrix}$$

Then

$$u_i(\beta_{\tau_1}^*, \beta_{\tau_2}^*) = -\begin{pmatrix}[I\{\varepsilon_{i,1} \le 0\} - \tau_1]\\ [I\{\varepsilon_{i,2} \le 0\} - \tau_2]\end{pmatrix} \otimes x_i.$$

Hence, $\mathrm{Var}(u_i(\beta_{\tau_1}^*, \beta_{\tau_2}^*)|\mathcal{F}_{i-1}) \to^P A \otimes \mathbb{E}x_i x_i'$, where

$$A := \begin{pmatrix}\tau_1(1-\tau_1) & \tau_1 \wedge \tau_2 - \tau_1\tau_2\\ \tau_1 \wedge \tau_2 - \tau_1\tau_2 & \tau_2(1-\tau_2).\end{pmatrix}.$$

Note that $\bar{H}u_j$ is an mds. We apply the mds CLT to the partial sum $\frac{1}{\sqrt{i}}\sum_{j=1}^{[ir]}\bar{H}u_j$, which converges weakly:

$$\sqrt{i}\bar{\Delta}_i(r) \Rightarrow \bar{Y}^{1/2}W(r), \quad r \in [0, 1]$$

where $W(r)$ stands for a vector of independent standard Wiener process on $[0, 1]$, and $\bar{Y} := \bar{H}(A \otimes \mathbb{E}x_i x_i')\bar{H}$. Now let $C_n(r) := G\sqrt{n}\bar{\Delta}_n(r)$, with $G = (I, -I)$. Also let $\Lambda = (G\bar{Y}G')^{1/2}$. Then for some vector of independent standard Wiener process $W^*(r)$, $C_n(r) \Rightarrow \Lambda W^*(r)$. In addition,

$$\begin{aligned} G\bar{V}_n G' = &= \frac{1}{n}\sum_{s=1}^n[C_n(\frac{s}{n}) - \frac{s}{n}C_n(1)][C_n(\frac{s}{n}) - \frac{s}{n}C_n(1)]'\\ &= \int_0^1[C_n(r) - rC_n(1)][C_n(r) - rC_n(1)]'dr,\end{aligned}$$

the last equality holds because $C_n(r)$ is a partial sum. Also, under the null that $\beta_{\tau_1,sub}^* = \beta_{\tau_2,sub}^*$,

$$\bar{\beta}_{n,sub}^1 - \bar{\beta}_{n,sub}^2 = G\begin{pmatrix}\bar{\beta}_{n,sub}^1 - \beta_{\tau_1,sub}^*\\ \bar{\beta}_{n,sub}^2 - \beta_{\tau_2,sub}^*\end{pmatrix} = \frac{1}{\sqrt{n}}C_n(1).$$

Hence the test statistic $n(\bar{\beta}_{n,sub}^1 - \bar{\beta}_{n,sub}^2)'(G\bar{V}_{n,sub}G')^{-1}(\bar{\beta}_{n,sub}^1 - \bar{\beta}_{n,sub}^2)$ equals

$$C_n(1)'\left(\int_0^1[C_n(r) - rC_n(1)][C_n(r) - rC_n(1)]'dr\right)^{-1}C_n(1)$$

is a continuous functional of $C_n(\cdot)$. By the continuous mapping theorem, the test statistic converges weakly to the desired limit. $\square$

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jeconom.2024.105673.

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